

Jeff H

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EDUCATION

Vellore Institute of Technology, Chennai
B.Tech in Computer Science And Engineering
St. Johns Public School
Grade XII

CGPA: 9.12
July 2024 – May 2028
Percentage: 92.2
2023 – 2024

TECHNICAL SKILLS

Languages: Java, Python, C++, JavaScript, SQL, C, HTML, CSS

Libraries & Frameworks: NumPy, Pandas, Scikit-learn, TensorFlow.js

Developer Tools: Git, GitHub, MySQL, Tableau, Octave

Core Concepts: Data Structures & Algorithms, Object-Oriented Programming (OOP), REST APIs, Machine Learning, Computer Vision

PROJECTS

SpotHole AI – AI-powered Road Hazard Detection Platform [GitHub](#)

Jan 2026 – Present

Computer Vision / Web Application

Python, YOLOv8, FastAPI, JavaScript

- Developed a YOLOv8-based real-time road hazard detection system using custom preprocessing and data augmentation techniques.
- Built a FastAPI backend exposing REST APIs for model inference and integrated it with a JavaScript-based frontend dashboard.
- Developed an interactive map interface to visualize detected potholes and generate route risk scores based on detection density.

Women's Health Analytics & Risk Prediction System [GitHub](#)

April 2026 – May 2026

Machine Learning / Data Analytics

Python, Scikit-learn, Tableau, Octave

- Integrated three healthcare datasets (maternal health, diabetes, lifestyle) into a unified dataset with preprocessing, feature engineering, and SMOTE for class imbalance.
- Built and compared five ML models (Logistic Regression, KNN, Decision Tree, Random Forest, SVM); SVM achieved the highest macro F1-score of 0.86 after hyperparameter tuning.
- Reduced dimensionality using PCA in Octave and built interactive Tableau dashboards to visualize health risk patterns across key features like BP, BMI, and blood sugar.

TurfArena – Sports Turf Booking Platform [GitHub](#)

Aug 2025 – Dec 2025

Full-Stack Web Development

Node.js, Express, MySQL, HTML, CSS, JavaScript

- Developed a full-stack turf booking platform with user authentication, slot availability management, and booking workflows.
- Modelled the database using ER diagrams and relational schema; implemented normalized MySQL tables with foreign key constraints for bookings, users, and turfs.
- Developed a RESTful backend with Node.js and Express, and connected it to a responsive HTML/CSS/JS frontend with dynamic slot rendering.

RESEARCH PUBLICATIONS

Alzheimer's Disease Detection Using Deep Learning Models [\[IEEE\]](#)

Dec 2025

International Conference on Intelligent Computing and Next Generation Networks (ICNGN 2025)

- Co-authored a research paper on detecting Alzheimer's disease using deep learning models applied to MRI brain scans.
- Evaluated CNN architectures including MobileNetV2, ResNet50, and InceptionV3 with augmented attention mechanisms.
- Compared MobileNetV2, ResNet50, and InceptionV3 for early-stage Alzheimer's disease detection using MRI scans.